

Lab E: Exp 22. The Suzuki Reaction

Mark Rejna

University of California, Berkeley

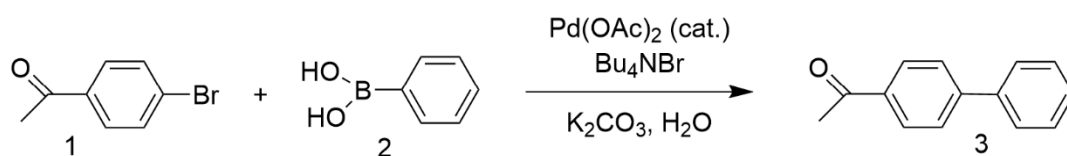
I. INTRODUCTION

The Suzuki reaction, developed in 1979 by Akira Suzuki and Norio Miyaura, is one of the most important organic chemistry advancements that won the 2010 Nobel Prize in Chemistry.¹ This reaction overcame the historical challenge of performing substitution reactions for nucleophilic addition on fully pi-conjugated aromatic rings by introducing a palladium catalysed cross-coupling reaction.

Similar carbon-carbon bond forming reactions, such as the Grignard reaction or the Negishi coupling reaction both use organometallic reagents, but they require strict anhydrous conditions. Research has shown that these reactions can be improved with the addition of several potential metallic catalysts which aim to improve yield and reaction optimisation.² In contrast, the Suzuki reaction is considered to be the simplest and most reliable method as it allows carbon-carbon bonds to form between a wide range of aryl halides and aryl boronic acids in more mild reaction conditions.³ Furthermore, the Suzuki reaction is highly advantageous as it is greener, due to its compatibility and use of water as a solvent, it is more selective as it leads to less side products, and it is more efficient as the palladium catalyst is regenerated.⁴

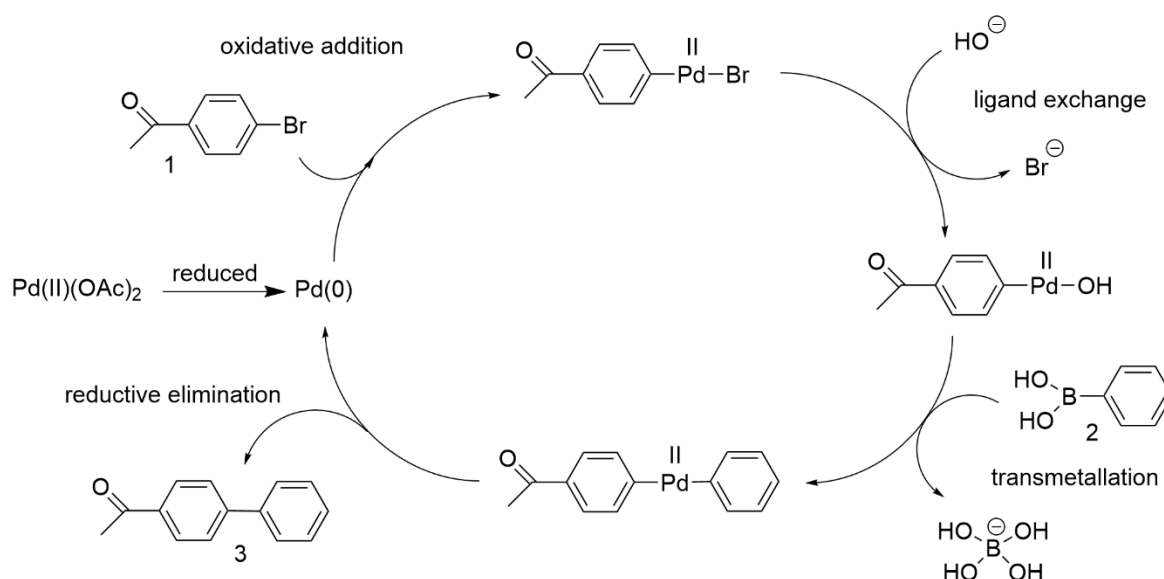
The specific Suzuki reaction performed in this investigation is the synthesis of 4-acetylbiphenyl (3) from 4-bromoacetophenone (1) and phenylboronic acid (2) in the presence of a palladium (II) acetate catalyst in basic aqueous conditions as outlined in *Scheme 1* below.

Scheme 1: General overview of the Pd-catalysed synthesis of 4-phenylbenzaldehyde



The reaction mechanism for the cross-coupling Suzuki reaction to form 4-acetylbiphenyl (3) is outlined below in *Scheme 2*.

Scheme 2: Catalytic cycle of 4-acetylbiphenyl synthesis



The palladium (II) acetate catalyst is reduced to palladium (0), where it undergoes oxidative addition with 4-bromoacetophenone (1) to form an organopalladium intermediate. A ligand exchange occurs due to the basic conditions and the presence of water, where the bromide ligand is replaced by a hydroxyl group. Next, transmetalation occurs as the phenyl group from phenylboronic acid (2) is transferred to the palladium centre, replacing the hydroxyl group. Finally, reductive elimination couples the two ligands to yield the product 4-acetylbiphenyl (3), which regenerates palladium (0).

II. EXPERIMENTAL PROCEDURE

This experiment was conducted by Mark Rejna, Ali Mozaffar, and Ibraheem Siddiqui following instructions outlined in the *Understanding the Principles of Organic Chemistry* textbook.⁵

4-Acetylbiphenyl (3)

4-bromoacetophenone (203 mg, 1.02 mmol), phenylboronic acid (134 mg, 1.10 mmol), palladium (II) acetate (2 mg, 0.01 mmol), tetrabutylammonium bromide (357 mg, 1.12 mmol), potassium carbonate (353 mg, 2.55 mmol), and distilled water (1 ml) were added to a long-neck 5 ml round bottom flask along with a magnetic stir bar. The flask was attached to a condenser with a gas trap apparatus. The reaction mixture was heated at reflux for 45 minutes at approximately 120 °C, the system was constantly stirred using the magnetic stir bar. The reaction mixture turned black and was left to cool to room temperature. The reaction mixture was then filtered on a Hirsch funnel leaving a grey solid that was washed with 10 ml hot water. The resulting solid was transferred to a centrifuge tube and acetone was added until the tube volume was equal to 3 ml. This was shaken vigorously until all solids dissolved, and then centrifuged for 2 minutes to separate the mixture into black sediment and a clear yellow supernatant that was extracted using a pipette. Water was added drop-wise to the supernatant, white precipitate formed, and the tube was shaken to redissolve the precipitate, this was repeated until crystals remained after being shaken, 5 water drops were added. The reaction mixture was then cooled to 0 °C in an ice bath and the dark yellow solution turned into a silver-white solid containing crystals. The resulting crystals were isolated on a Hirsch funnel, washed with an ice cold water, and dried by drawing a vacuum through the filtrate. The yield of the resulting crystal product, 4-phenylbenzaldehyde, was measured (185.6 mg, 93 %). The melting range of the product was measured to be between 124 and 125.6 °C.

^1H NMR (400 MHz, CDCl_3) δ 8.0 (d, J = 6-10 Hz, 2H), 7.5-7.7 (m, 4H), 7.2-7.4 (m, 3H), 2.5 (s, 3H)

^{13}C NMR (100 MHz, CDCl_3) δ 198.5, 156.2, 140.0, 137.1, 128.9, 128.3, 127.3, 28.5

III. DISCUSSION

RESULTS AND DATA ANALYSIS

The mass recovered of the clear crystal product 4-acetylbiphenyl was measured to be 185.6 mg which represents a yield of 93 %. The melting range of this product was measured to be between 124 and 125.6 °C.

A potential reason that the high yield of 93 % is less than 100 % is because some of the product was lost as it could not be removed from the long-neck 5 ml round bottom flask due to the flasks long-neck and spherical shape as well as the product sticking to the flask walls. To remove this systematic error, 1 to 2 ml of solvent should be added to the flask, the mixture should be gently heated, and then vigorously shaken to free the trapped solid.

Another potential cause decreasing the yield of the product is that some of the palladium (II) acetate was lost and not transferred into the reaction flask as it was a fine powder solid. To eliminate this transfer loss, palladium (II) acetate should be thoroughly transferred to the reaction flask.

The melting range of the product, 4-acetylbiphenyl was measured to be between 124 and 125.6 °C. The literature melting range of the product is between 117 and 124 °C. The observed melting range is within the literature range, therefore, it can be determined that the product is pure. Potential reasons for the observed melting range to exceed slightly higher than the literature range include the melting point apparatus heating too quickly and causing the thermometer to show a higher temperature than the temperature of the heating block, the

thermometer could have been calibrated incorrectly, or this could be due to human error in recording the measurement. To remove this error the thermometer should be calibrated before use, the melting point apparatus should increase temperature at a slow rate for an accurate measurement, and measurements should be taken precisely.

Overall, by implementing the suggested improvements of the errors listed above, the yield of both products can be improved.

IV. REFERENCES

1. Nobel Prize Outreach AB. The Nobel Prize in Chemistry 2010: Illustrated Information. NobelPrize.org. <https://www.nobelprize.org/prizes/chemistry/2010/illustrated-information/> (accessed Mar 18, 2026).
2. Haas, D.; Hammann, J. M.; Greiner, R.; Knochel, P. Recent Developments in Negishi Cross-Coupling Reactions. *ACS Catal.* 2016, 6 (3), 1540–1552. <https://doi.org/10.1021/acscatal.5b02718>
3. Li, B.; Zeng, H. C. Minimalization of Metallic Pd Formation in Suzuki Reaction with a Solid-State Organometallic Catalyst. *ACS Appl. Mater. Interfaces* 2020, 12 (30), 33671–33681. <https://doi.org/10.1021/acsami.0c09739>
4. Pasricha, S.; Sachidanand; Srivastava, A.; Yadav, S.; Tuwani, N.; Rangarajan, T. M.; Mittal, K. Greener Media for Nano Catalysts in Suzuki Miyaura Reaction. *Coord. Chem. Rev.* 2025, 528, No. 216431. <https://doi.org/10.1016/j.ccr.2025.216431>
5. Pedersen, S. F.; Myers, A. M. *Understanding the Principles of Organic Chemistry: A Laboratory Course*, 1st ed.; Cengage Learning, 2011.